Mastering Git & GitHub

The Essential Toolkit for Modern Developers

Version Control, Collaboration, and Open Source Basics

Why Version Control?

Track Every Change

Maintain a complete history of your project. Never lose a line of code again. Go back in time to any version instantly.

Collaborate Safely

Work on the same codebase with thousands of developers. Merge changes without overwriting each other's hard work.

Git vs. GitHub

- **Git:** The local tool. It lives on your computer and manages your personal history.
- **GitHub:** The cloud platform. It hosts your Git repositories so others can access them.
- **GitHub** adds social features: Pull Requests, Issues, and Actions.

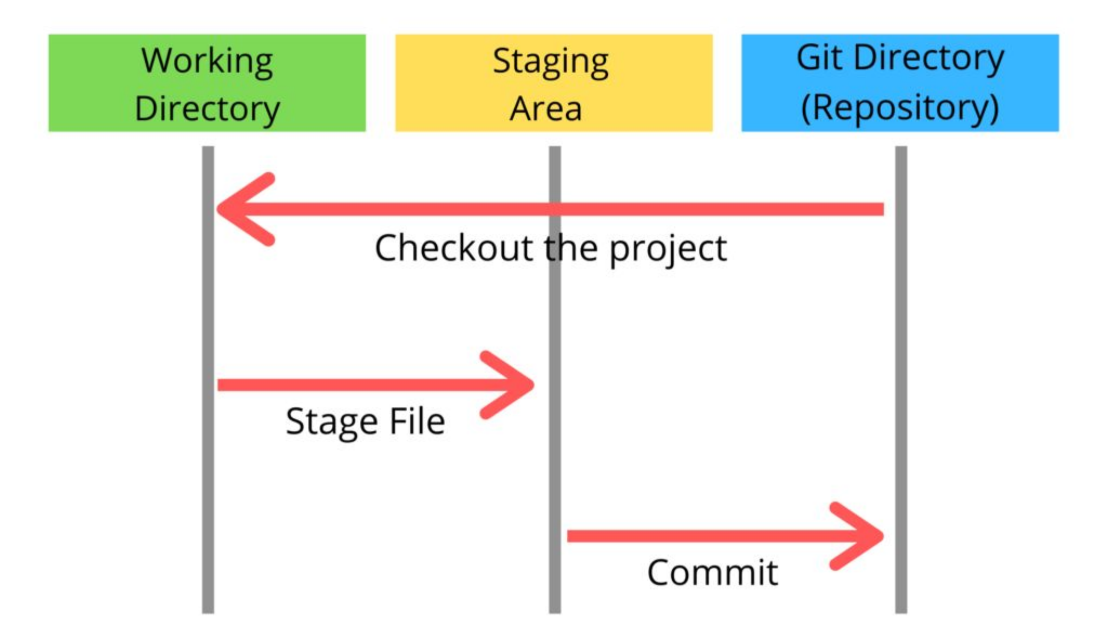
The Three Stages of Git

Understanding how files move through Git is key to mastering the workflow:

Working Directory: Untracked files or modified changes.

Staging Area: Files prepped for the next snapshot.

Repository: Permanently stored snapshots (Commits).



Essential Setup

- **Initialize:** git init creates a new local repository in your current folder.
- **Identity:** git config --global user.name "Your Name"
- **Email:** git config --global user.email "email@example.com"
- **Status:** git status is your best friend. Use it to see where your files are.

The Local Workflow

Step Process to Save

1. **Modify:** Edit your files in your editor.
2. **Stage:** Use `git add [filename]` to prep changes.
3. **Commit:** Use `git commit -m "Message"` to save forever.

Think of it like taking a photo: Frame the shot (Add), then click the button (Commit).

Branching & Merging

Feature Branches

Never work directly on the main branch. Create a new branch for every task using `git checkout -b feature-name`

Merging Back

When the feature is done, switch back to main and bring the changes in: `git merge feature-name`

Entering GitHub

Collaboration at Scale

GitHub provides a central "Remote" for your local "Local" repository. It enables:

- Forking other projects
- Bookmarking repositories
- Organization management

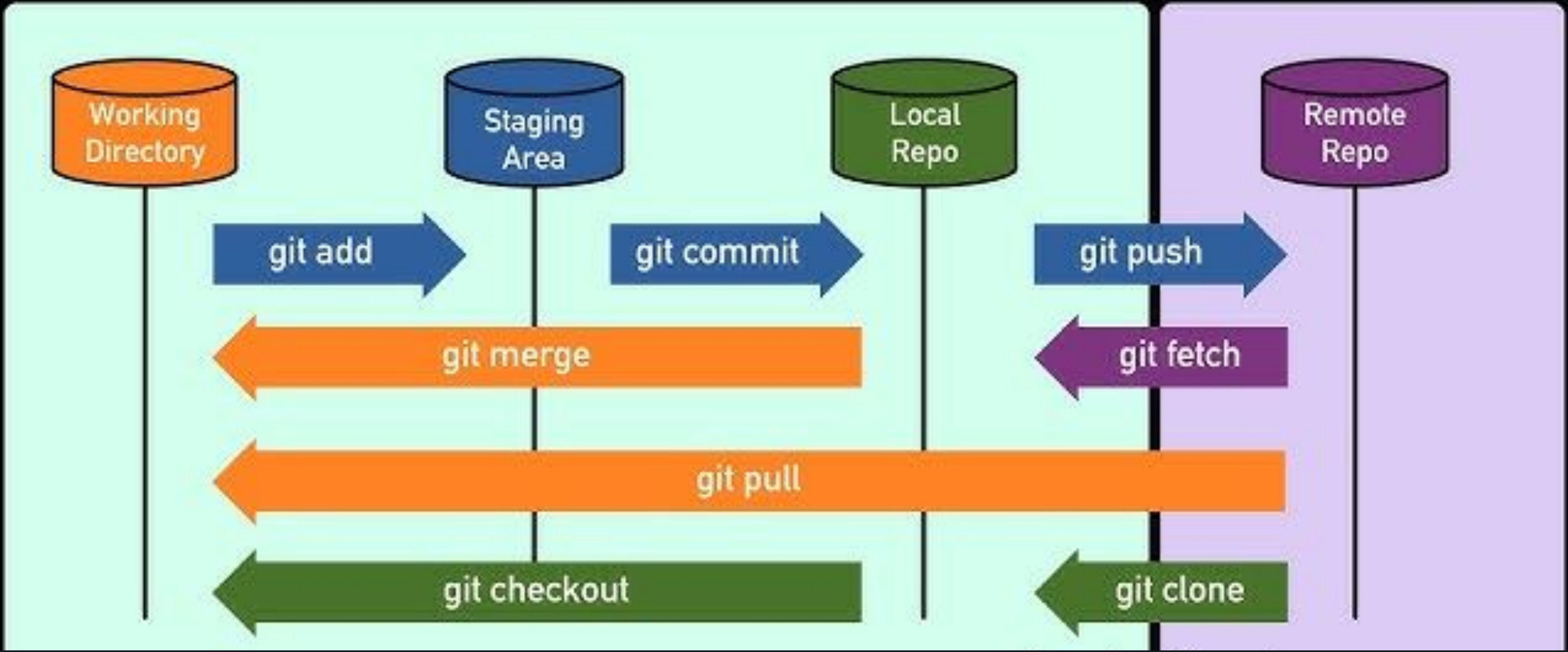
GitHub Pull Request Discussion Interface

The Pull Request

A **Pull Request (PR)** is how you propose changes on GitHub. It allows for team discussion and code review before merging.

1. Push feature branch to GitHub
2. Open a Pull Request via UI
3. Teammates review & comment
4. Merge into Main!

How Github Actually Works?



Syncing with Remote

Command	Action	Direction
git push	Sends local commits to GitHub	Local → Remote
git pull	Fetches and merges remote changes	Remote → Local
git fetch	Downloads changes (no merging)	Remote → Local
git clone	Creates a local copy of a repo	Remote → Local

Handling Conflicts

What is a Conflict?

When two people change the **same line** of the **same file** differently, Git doesn't know which one to keep.

The Solution

1. Open the file.
2. Delete the markers (<<<<).
3. Choose the code to keep.
4. Add and commit the fix.

Questions?

Thank you for joining the session!

Happy Coding & Versioning.

Refer to this link for doubts later:

https://www.w3schools.com/git/default.asp?remote=github